CHAPTER 6
SENSOR AND TRANSDUCER

6.1 Introduction

An important function in the industrial and scientific use of electronics is the measurement of physical parameters such as position, temperature, force, pressure, and the rate of flow of fluids. Without transducers, advances in the application of control and computation would have been impossible.

6.2 Definition of a transducer

Transducer is any device that converts energy in one form to energy in another. The majority either convert electrical energy to mechanical displacement or convert some *nonelectrical* physical quantity, such as temperature, sound or light to an electrical signal.

The functions of a transducer are:

1) to sense the presence, magnitude, change in, and frequency some of measurand
2) to provide an electrical output that, when appropriately processed and applied to readout device, gives accurate quantitative data about the measurand (Figure 6.1).

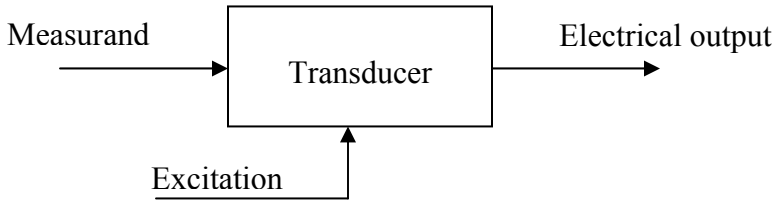


Figure 6.1: Block diagram of a transducer

The term measurand refers to the quantity, property, or condition which the transducer translates to an electrical signal.

6.3 Classification of transducers

Transducer can be classified according to their application, based primarily on the physical quantity, property, or condition that is measured.

The transducer can be categories into:

1) Passive transducer: these transducers require an external power, and their output is a measure of some variation, such resistance and capacitance.
2) Self generating transducer: These transducers do not require an external power, and they produce analog voltage or current when stimulated by some physical form of energy.

6.4 Selecting a transducer

In the selection of a transducer, there are approximately eight areas of concentration.

1. Operating range: The transducer should maintain range requirements and good resolution.
2. Sensitivity: the transducer must be sensitive enough to allow sufficient output.
3. Frequency response and resonant frequency
4. Environmental compatibility: Do the temperature range of the transducer its corrosives fluids, the pressures, shocks, and interactions it is subject to its size and mounting restrictions make it inapplicable.
5. Minimum sensitivity: The transducer must be minimally sensitivity to expected stimuli other than the measurand.
6. Accuracy: The transducer may be subject to repeatability and calibration errors as well as error expected owing to sensitivity to other stimuli.
7. Usage and ruggedness. The ruggedness both of mechanical and electrical intensities versus size and weight must be considered
8. Electrical parameter. Length and type of cable required, signal to noise ratio (SNR) when combined with amplifiers and frequency response limitations.

6.5 Temperature transducers

Temperature transducers can be divided into four main categories:

1. Resistance temperature detectors (RTD)
2. Thermocouples
3. Thermistor
4. Ultrasonic transducers

6.5.1 Resistance temperature detectors (RTD)

Detectors of wire resistance temperature common employ platinum, nickel or resistance wire elements, whose resistance variation with temperature has high intrinsic accuracy. They are available in many configurations and size and as shielded or open units for both immersion and surface applications.

The relationship between temperature and resistance of conductors can be calculated from the equation:

$$R = R_0(1 + \alpha\Delta T)$$

where

R = the resistance of the conductor at temperature t (^{0}C)

R_0 = the resistance at the reference temperature, usually 20^{0}C

α = the temperature coefficient of resistance

ΔT = the difference between the operating and the reference temperature

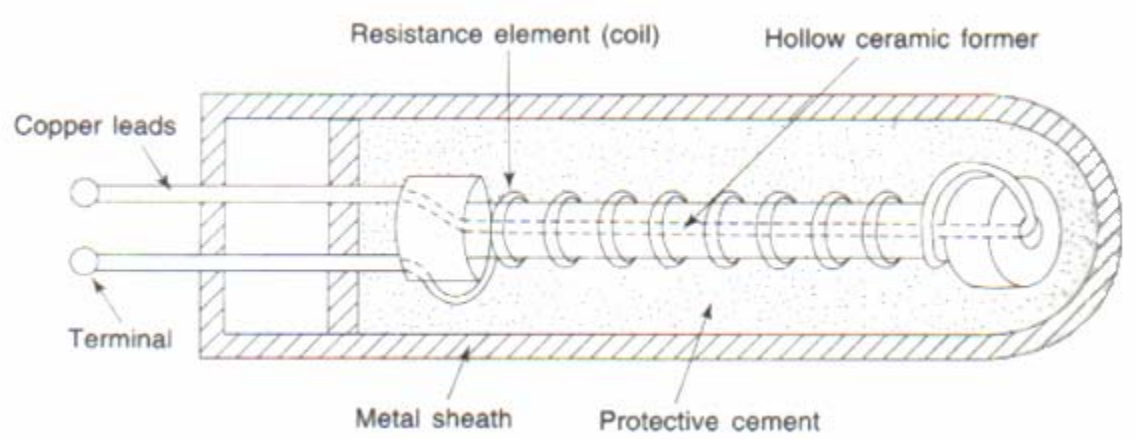


Figure 6.2: Construction of a platinum resistance temperature detector

Example:

A platinum resistance thermometer has a resistance of 150Ω at 20^0C. Calculate its resistance at 50^0C ($\alpha_{20} = 0.00392$)

$$R = R_0(1 + \alpha\Delta T)$$
$$= 150\ (1 + 0.00392(50 - 20))$$
$$= 167.64\Omega$$

6.5.2 Thermocouples

The thermocouple is a temperature transducer that develops an emf which is a function of the temperature between hot junction and cold junction. The construction of a thermocouple is quite simple. It consists of two wires of different metals are joined together at one end, a temperature difference between this end and the other end of wires produces a voltage between the wires (Figure 6.3). The magnitude of this voltage depends on the materials used for the wires and the amount of temperature difference between the joined ends and the other ends.

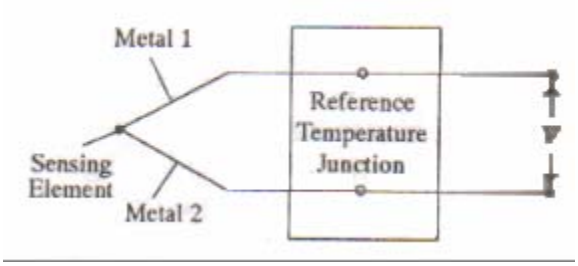


Figure 6.3: The basic thermocouple assembly

The junction of the two wires of the thermocouple is called the sensing junction. In normal use this junction is placed in or on the material being tested, and the other ends of the wire are connected to the voltage measuring equipment. Since the temperature difference between this sensing junction and the other ends is the critical factor, the other ends are either kept at a constant reference temperature or, when the cost of the equipment is very low, simply maintained at room temperature. When the other ends are kept at room temperature, the temperature is monitored and the thermocouple output voltage readings are corrected any changes in room temperature. It is because the temperature at this end of the thermocouple wires is a reference temperature, the junction here with the equipment terminals or with other connecting wires is known as the reference, also called as the cold junction.

When a temperature difference exists between the sensing junction and the reference junction, an emf is produced, which causes a current in the circuit. When the reference end is terminated by meter or recording device, the meter indication will be proportional to the temperature difference between the hot junction and the reference junction. This thermoelectric effect, caused by contact potential at the junctions, is known as Seebeck effect.

The magnitude of the thermal emfs depend on the wire materials used, and on the the temperature difference between the junctions.

The emf of the thermocouple is given as

$$E = c(T_1 - T_2) + k(T_1^2 - T_2^2)$$

Where

c and k = constant of the thermocouple materials

T_1 = The temperature of the "hot" junction

T_2 = The temperature of the "cold" or "reference" junction

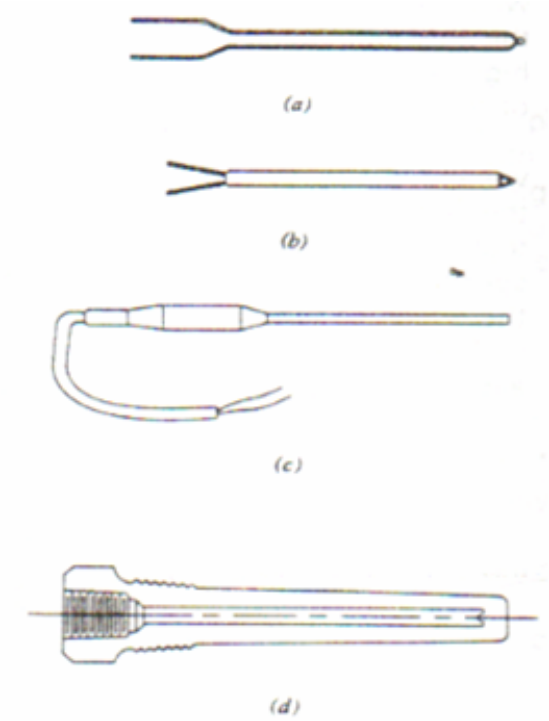


Figure 6.4: Thermocouples and thermocouple assemblies: a) Uninsulated thermocouple b) insulated thermocouple c) probe assembly d) thermocouple well

Advantages of thermocouples:

1. It has rugged construction
2. Temperature range -270^0C to 2700^0C
3. Using extension leads and compensating cables, long transmission distances for temperature measurement are possible
4. Bridge circuits are not needed for temperature measurement
5. Comparatively cheaper in cost
6. Calibration checks can be easily performed
7. Thermocouples offer good reproducibility

8. Speed of response is high compared to the filled system thermometer
9. Measurement accuracy is quite good

Limitations of thermocouples:

1. Cold junction and other compensation is essential for accurate temperature measurements
2. They exhibit non linearity in the emf vs temperature characteristics
3. Stray voltage pick up may be possible
4. To avoid stray electrical signal pick up, proper separation of extension leads from thermocouple wire is essential
5. In many applications, the signal needs to amplified

Example:

During experiments with a copper-constantan thermocouple it was found that $c = 3.75 \times 10^{-2}$ mV/^{0}C and $k = 4.5 \times 10^{-5}$ mV/^{0}C^2. If $T_1 = 100^0$C and the cold junction T_2 is kept in ice, compute the resultant electromotive force.

6.5.3 Thermistors

The electrical resistance of most materials changes with the temperature. By choosing materials that are useful in temperature control circuits as well as in temperature measurement. A thermistor is a semiconductor made by sintering mixtures of metallic oxide, such as oxides of manganese, nickel, cobalt, copper and uranium.

Thermistors have negative temperature coefficient. That is, their resistance decreases as their temperature rises.

Resistance at 25^0C for typical commercial unit ranges from the vicinity of 100Ω to over 10MΩ. Figure 5 shows resistance versus temperature for a family thermistor. The resistance value marked at the bottom end of each curve is a value at 25^0C. The smallest thermistors are made in the form of beads. Where greater power dissipations is required, thermistors may be obtained in disc, washer or rod.

Types of thermistor	Resistance
Disc	1 to 1MΩ
Washer	1 to 50kΩ
Rod	high resistance

Thermistors can be connected in series or parallel combinations for applications. Typical thermistor configurations are shown in Figure 6.6. Thermistor are non linear devices over temperature range, although now units with better than 0.2% linearity over the 0 - 100^0C temperature range are available.

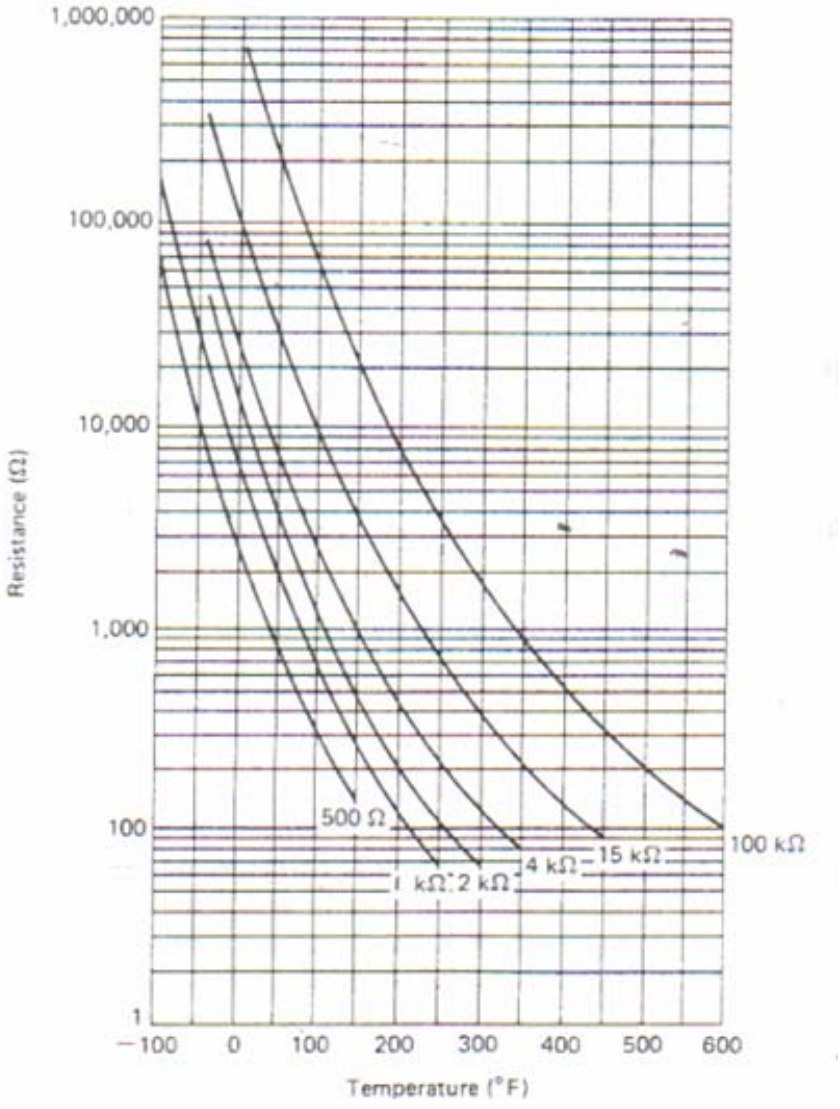


Figure 6.5: Typical thermistor resistance versus temperature curves

Advantage of Thermistor

1. Small size and low cost
2. Fast response over narrow temperature range
3. Good sensitivity in Negative Temperature Coefficient (NTC) region
4. Cold junction compensation not required due to dependence of resistance on absolute temperature.
5. Contact and lead resistance problems not encountered due to large resistance

Limitations of thermistor

1. Non linearity in resistance vs temperature characteristics
2. Unsuitable for wide temperature range
3. Very low excitation current to avoids self heating
4. Need of shielded power lines, filters, etc due to high resistance.

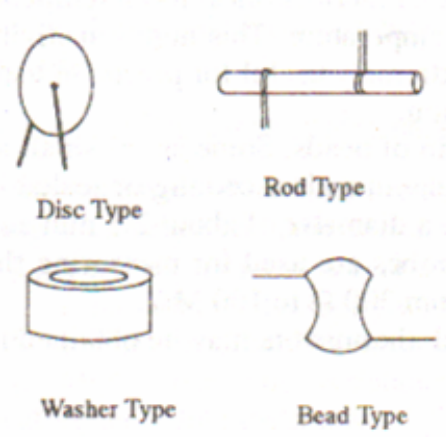


Figure 6.6: Various configurations of thermistor

Example

The circuit of Figure 6.7 is to be used for temperature measurement. The thermistor is a 4kΩ type identified in Figure 6.5. The meter is a 50mA ammeter with a resistance of 3Ω. Rc is set to 17Ω, and the supply voltage V_T is 15V. What will the meter readings at 77^0F and 150^0F be?

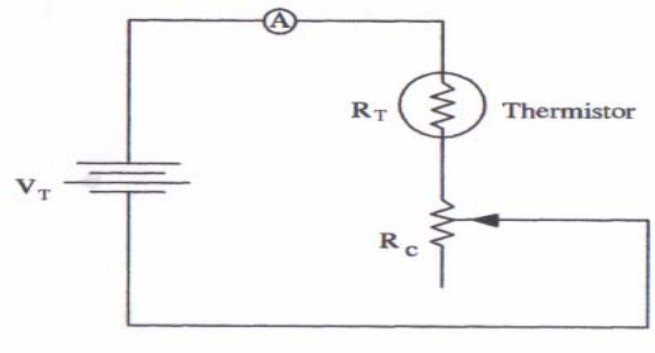


Figure 6.7: Basic thermistor circuit for measuring

Solution:

6.6 Resistance Position Transducer

The principle of the resistance transducer is that the physical variable under measurement causes a resistance change in the sensing element. A common requirement in industrial measurement and control work is to be able to sense the position of an object or distance

it has moved. One type of displacement transducer uses a resistance element with a sliding contact or wiper linked to the object being monitored. Thus, the resistance between the slider and one end of the resistance element depends on the position of the object. Figure 6.8 shows the construction of this type of transducer.

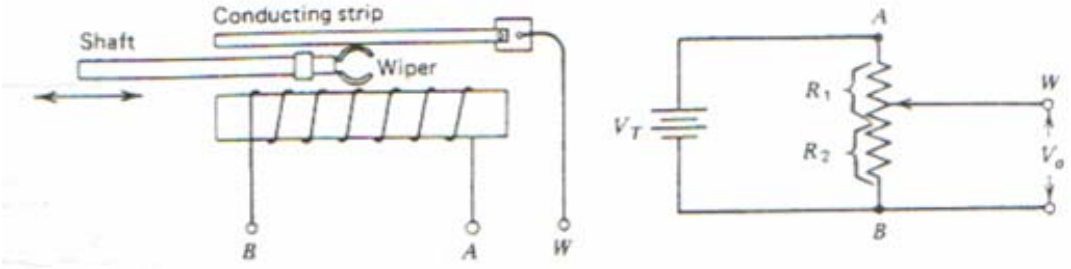


Figure 6.8: Resistance position transducer or displacement transducer

The output voltage depends on the wiper position and therefore is a function of the shaft position. This voltage may be applied to a voltmeter calibrated in inches for visual display.

Consider to Figure 6.8, if the circuit is unloaded, the output voltage is a certain fraction of V_T, depending on the position of the wiper:

$$\frac{V_o}{V_T} = \frac{R_2}{R_1 + R_2}$$

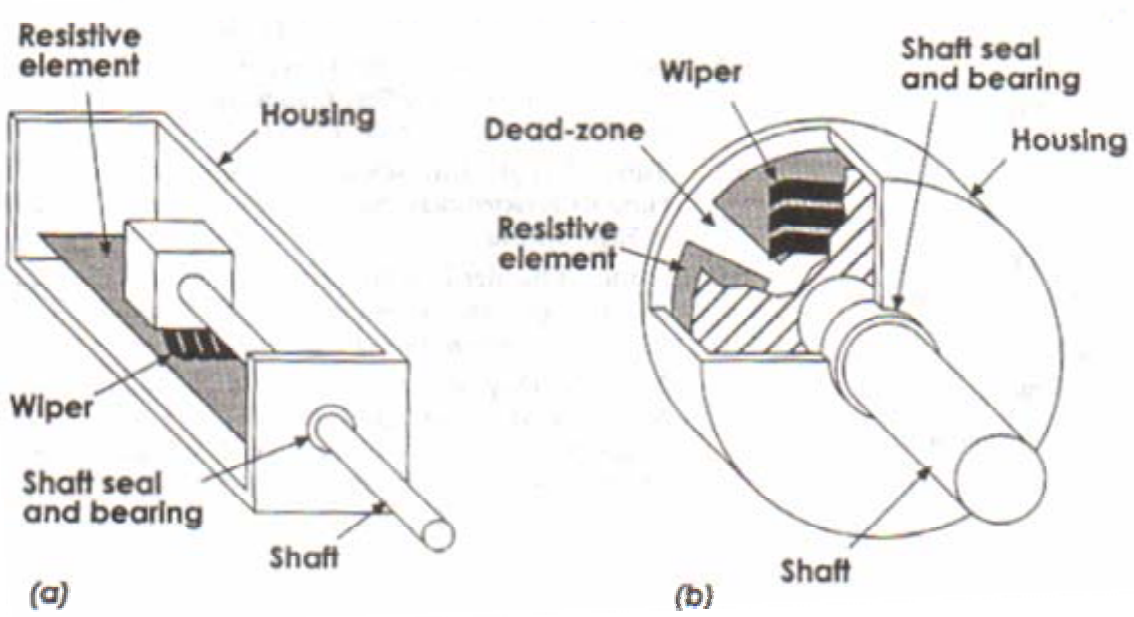


Figure 6.9: Representative cutaways of linear motion: a) rotary b) potentiometer

Example:

A displacement transducer with a shaft stroke of 3.0in is applied to the circuit of Figure 6.8. The total resistance of potentiometer is 5kΩ, and the applied voltage V_T = 5V. When the wiper is 9in from B, what is the value of the output voltage V_O?

Solution:

Example:

A resistive position transducer with a resistance of 5000Ω and shaft stroke of 5in is used in the arrangement of Figure 6.10. Potentiometer R_3R_4 is also 5000Ω and V_T = 5V. The initial position to be used as a reference point is such that $R_1 = R_2$ (i.e., the shaft is at mid stroke). At the start of the test, potentiometer R_3R_4 is adjusted so that the bridge is balanced (V_E = 0). Assuming that the object being monitored will move a maximum distance of 0.5in toward A, what will the new value of V_E be?

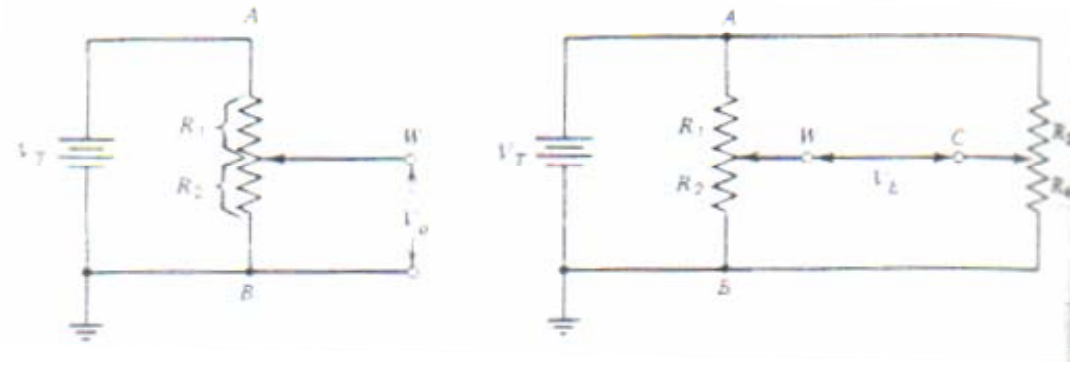


Figure 6.10: Basic voltage divider and resistance bridge circuits

Solution:

6.7 Capacitive Transducers

The capacitance of a parallel plate capacitor is given by

$$C = \frac{kA\varepsilon_0}{d}(Farads)$$

where

k = dielectric constant

A = the area of the plate, in m^2

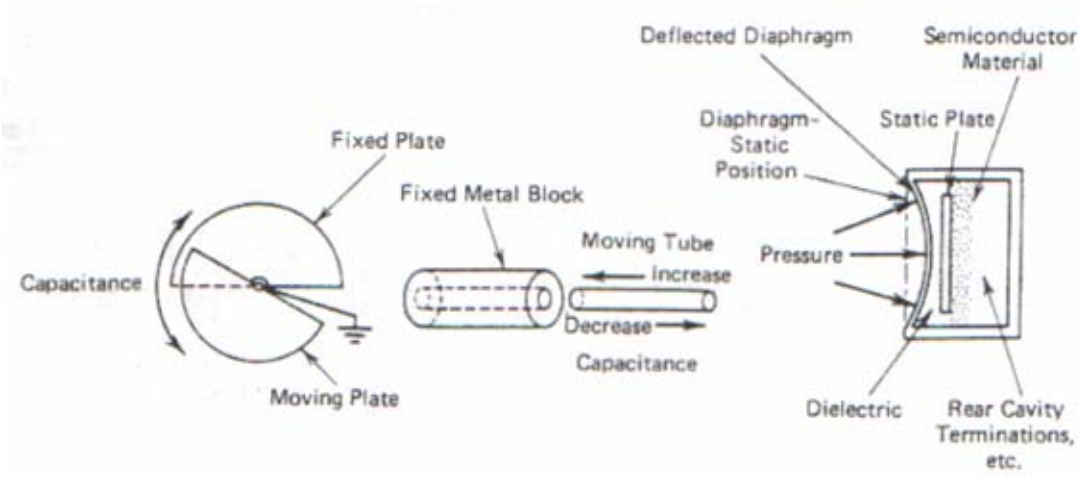
ε_0 = 8.854 x 10^{-12} F/m

d = the plate placing in m

since the capacitance is inversely proportional to the spacing of the parallel plates, any variation in d causes a corresponding variation in the capacitance.

Figure 6.11 shows several forms of capacitive transducers such as:

1. Rotary plate capacitor (Figure 6.11-a): The capacitance of this unit proportional to the amount of the fixed plate that is covered, that shaded by moving plate. This type of transducer will give sign proportional to curvilinear displacement or angular velocity.
2. Rectilinear capacitance transducer (Figure 6.11-b): It consists of a fixed cylinder and a moving cylinder. These pieces are configured so the moving piece fits inside the fixed piece but insulated from it.
3. Thin diaphragm (Figure 6.11-c): A transducer that varies the spacing between surfaces. The dielectric is either air or vacuum.



a) b) c)

Figure 6.11: Capacitance transducers

Advantages:

1. Has excellent frequency response
2. Can measure both static and dynamic phenomena.

Disadvantages:

1. Sensitivity to temperature variations
2. the possibility of erratic or distortion signals owing to long lead length

Applications:

1. As frequency modulator in RF oscillator
2. In capacitance microphone
3. Use the capacitance transducer in an ac bridge circuit

Example:

An electrode diaphragm pressure transducer has plates whose are is $5 \times 10^{-3} m^2$ and whose distance between plates is $1 \times 10^{-3} m$. Calculate its capacitance if it measures air pressure. The dielectric constant of air is 1.

Solution:

6.8 Inductive transducer

Inductive transducers may be either of the self generating or passive type. The self generating type utilises the basic electrical generator principle, i.e, a motion between a conductor and magnetic field induces a voltage in the conductor (generator action). This relative motion between the field and the conductor is supplied by changes in the measurand.

An inductive electromechanical transducer is a device that converts physical motion (position change) into a change in inductance. Transducers of variable inductance type work upon one of the following principles:

1. Variation of self inductance
2. Variation of mutual inductance

Inductive transducers are mainly used for the measurement of displacement. The displacement to be measured is arranged to cause variation in any of three variables:

1. Number of turns
2. Geometric configuration
3. Permeability of the magnetic material or magnetic circuits

For example, let us consider the case of a general inductive transducer. The inductive transducer has N turns and reluctance R. When a current i is passed through it, the flux is

$$\phi = \frac{Ni}{R}$$

Therefore $$\frac{d\phi}{dt} = \frac{N}{R} x \frac{di}{dt} - \frac{Ni}{R^2} x \frac{dR}{dt}$$

If the current varies very rapidly,

$$\frac{d\phi}{dt} = \frac{N}{R} x \frac{di}{dt}$$

But emf induced in the coil is given by e = N x dΦ/dt

Therefore $$e = Nx \frac{N}{R} x \frac{di}{dt} = \frac{N^2}{R} x \frac{di}{dt}$$

Also the self inductance is given by

$$L = \frac{e}{\frac{di}{dt}} = \frac{N^2}{R}$$

Therefore, the output from an inductive transducer can be in the form of either a change in voltage or a change in inductance.

6.8.1 Change in self inductance with number of turns

The output may be caused by a changed in the number of turns. Figure 6.12a and 6.12b are transducers used for the measurement of displacement of linear and angular movement respectively.

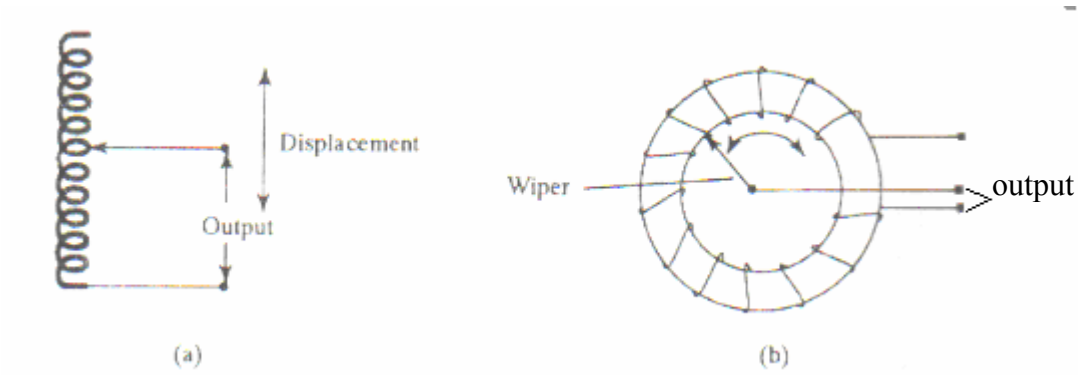


Figure 6.12

Figure 6.12(a) is an air cored transducer for measurement of linear displacement. Figure 6.12(b) is an iron cored coil used for the measurement of angular displacement. In both cases, as the numbers of turns are changed, the self inductance and the output also changed.

6.8.2 Transducer working on the principle of change in self inductance with change in permeability

Figure 6.13 shows an inductive transducer with works on the principle of the variation of permeability causing a change in self inductance. The iron core is surrounded by the winding. If the core is inside the winding, its permeability is increased, and so is the inductance. When the iron core is moved out of the winding, the permeability will decreases, resulting in a reduction of the self inductance on the coil. This transducer can be used for measuring displacement.

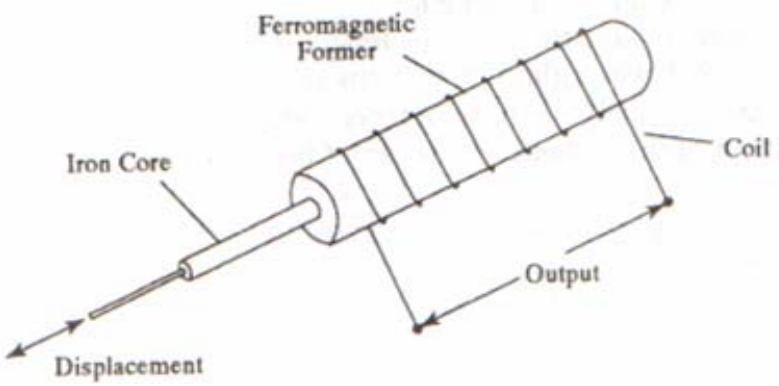


Figure 6.13: Inductive transducer working on the principle of variation of permeability

6.9 Strain Gauge

The strain gauge is an example of a passive transducer that uses electric resistance variation in wires to sense the strain produced by a force on wires. It is a very versatile detector and transducer for measuring weight, pressure, mechanical force, or displacement.

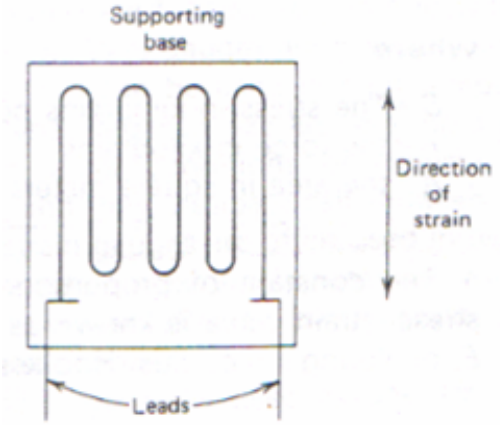


Figure 6.14: Resistive strain gauges: wire construction

The construction of a bonded strain gauge (Figure 6.14) shows a fine wire element looped back and forth on a mounting plate, which is usually cemented to the member undergoing stress. A tensile stress tends to elongate the wire and thereby increase its

length and decrease its cross-sectional area. The combined effect is an increase in resistance as seen from equation (1)

$$R = \frac{\rho L}{A} \qquad (1)$$

Where

ρ = the specific resistance of the conductor material in ohm meters

L = the length of the conductor in meters

A = the area of the conductor in square meters

As a consequence of strain, two physical qualities are of particular interest:

1. the change in gauge resistance
2. the change in length

The relationship between these two variables expressed as a ratio is called the gauge factor, K

$$K = \frac{\frac{\Delta R}{R}}{\frac{\Delta L}{L}} \qquad (2)$$

Where

K = the gauge factor

R = the initial resistance in ohms (without strain)

ΔR = the change in initial resistance in ohms

L = the initial length in meters (without strain)

ΔL = the change in initial length in meters

The term $\frac{\Delta L}{L}$ in the denominator is the same as the unit strain G. Therefore, equation (2) can be written as

$$K = \frac{\frac{\Delta R}{R}}{G} \qquad (3)$$

Stress is defined as the internal force per unit area. The stress equation is

$$S = \frac{F}{A} \qquad (4)$$

Where

S = the stress (kg/m^2)

F = the force (kg)

A = the area (m^2)

The constant of proportionality between stress and strain for a linear stress-strain curve is known as the modulus of elasticity of the material E, or Young's modulus. Hooke's law is written as

$$E = \frac{S}{G}$$

Where

E = Young's modulus (kg/m^2)

S = the stress (kg)

G = the strain (no units)

Example:

A resistance strain gauge with a gauge factor of 2 is fastened to a steel member, which is subjected to a strain of 1×10^{-6}. If the original resistance value of the gauge is 130Ω, calculate the change in resistance.

Example:

A round steel bar, 0.02m in diameter and 0.40m in length, is subjected a tensile force of 33000 kg, where $E = 2 \times 10^{10} kg/m^2$. Calculate the elongation, ΔL, in meters.

6.10 Linear variable differential transformer (LVDT)

It consists basically of a primary winding and two secondary windings, wound over a hollow tube and positioned so the primary winding is between two secondaries. Figure 6.15 shows the construction of the LVDT. An iron core slides within the tube and therefore affects the magnet coupling between the primary and the two secondaries. When the core is in the centre, voltage induced in the two secondaries is equal. When the core is moved in one direction from centre, the voltage induced in one winding is increased and that in the other is decreased. Movement in the opposite direction reverses this effect.

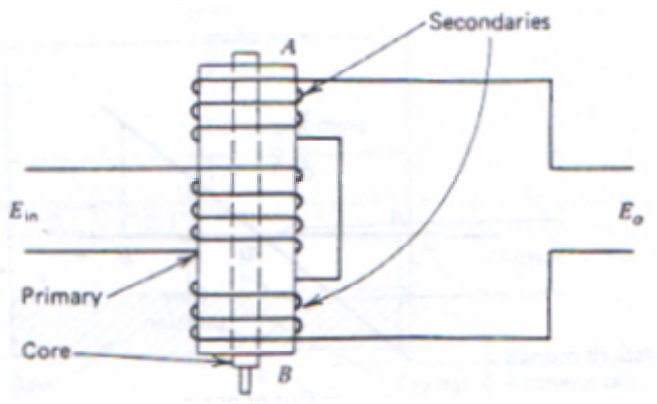


Figure 6.15: The construction of linear variable differential transformer

In the schematic diagram shown in Figure 6.16, the windings are connected "series opposite". That is, the polarities of V_1 and V_2 oppose each other as we trace through the circuit from terminal A to terminal B. Consequently, when the core is in the centre so that $V_1 = V_2$, there is no voltage output, $V_o = 0$.

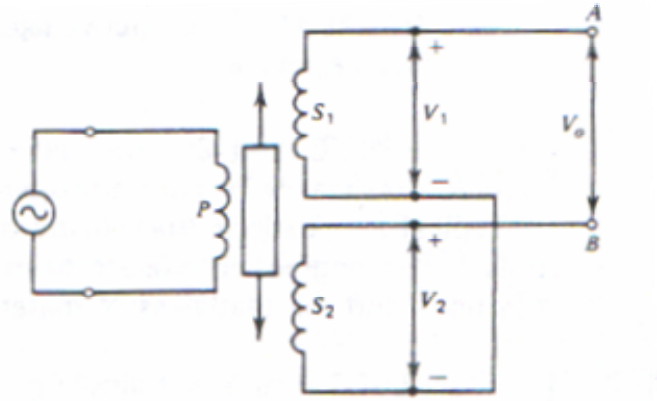
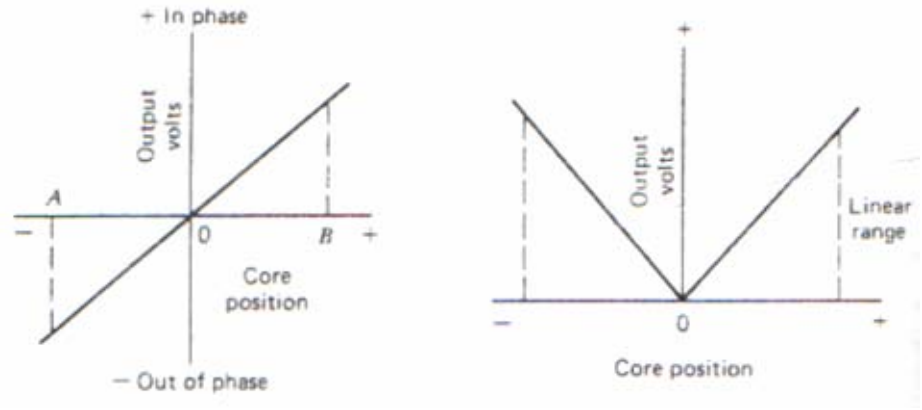


Figure 6.16: Schematic diagram

When the core is away from the centre toward S_1, V_1 is greater than V_2 and the output voltage V_o will have polarity of V_1, When the core is away from centre toward S_2, V_2 is greater than V_1 and the output will have the polarity of V_2. That is, the output ac voltage inverts as the core passes the centre position. The farther the core moves from centre, the greater the difference in value betweenV_1 and V_2, and consequently the greater the value of V_o. Thus, the amplitudes of V_o is a function of the distance the core has moved, and the

polarity or phase indicates which direction it has moved (Figure 6.17). If the core attached to a moving object, the LVDT output voltage can be a measure of the position of the object.



a) b)

Figure 6.17: Output voltage a) phase relationship b) absolute magnitude

Example:

An ac LVDT ha s the following data: input 6.3V, output 5.2V range ±0.5in. Determine

a) The plot of the output voltage versus core position for a core movement going from +0.45 to -0.03in.
b) The output voltage when the core is -0.25in from centre